

Carter Tull

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Software Engineer/Automation Developer

Agile Software Development | Full Stack Development | Application Architecture

Forward-thinking software engineering professional specializing in automation development with hands-on experience in API integration and data transformation. Known for designing TypeScript-based automation solutions that streamline business processes and implement intelligent data matching systems. **Successes include engineering microservices for automated construction project data processing and building API aggregators to accelerate bid visibility.** Previous experience includes optimizing SQL queries, integrating NgRx into Angular systems, and contributing to test automation within Agile environments. Strong background in full stack development with training in data structures, algorithms, and cloud services. **Earned rank of Eagle Scout by leading large-scale community service initiatives and demonstrating sustained commitment to leadership, project execution, and civic engagement.**

Core Competencies

- Object-Oriented Programming
- Frontend/Backend Development
- Cloud-Based Deployment
- RESTful API Integration
- Version Control
- Software Debugging
- Test Case Development
- Source Code Management
- Algorithm Design
- Project Lifecycle Management
- User Interface Design
- Cross-Platform Development
- Code Review Processes
- Scrum Methodology
- Responsive Design

Education

Bachelor of Science (BS), Computer & Information Science

The Ohio State University – Columbus, OH | 2025

- Relevant Coursework: Software Applications Capstone, Programming Languages, Software I and II, Foundations I and II, Systems I and II, Web Applications Project Course, Computer Networking, Computer Game Animation, Data Management in the Cloud, Software Engineering Techniques, Computer Architecture, UNIX Programming Environment, Java, Database Systems, Information Security, Digital Logic, Calc 1, Calc 2, Probability and Statistics for Engineers, Foundations of Higher Mathematics, Physics (Mechanical, Energy, Thermodynamic), Ethics in the Professions, Public Speaking
- The Ohio State University Marching Band – 4-year member (scholarship), Summer Clinic Counselor, Summer Sessions Marching Team Member, Tuba Fours (2 years), 3-time Script Ohio i-Dotter
- Member - Ohio State Competitive Programming Club

Select Projects

Quandary Interpreter | Java, JFlex, Java CUP, Git (January 2025-April 2025)

- Implemented full-stack interpreter for Quandary programming language blending functional and imperative paradigms.
- Extended lexical analyzer using JFlex and designed context-free grammar with Java CUP to construct abstract syntax trees.
- Built AST node class hierarchy using visitor design pattern to support interpreter traversal and evaluation.
- Developed interpreter with support for recursive functions, concurrency, and mark-sweep garbage collection.

E-Commission Card | JavaScript, React Native, Expo, SQL, AWS (January 2025-April 2025)

- Led development of capstone project for NBBi to streamline E-Commission card management.
- Engineered cross-platform mobile application using React Native with Expo for front end.
- Utilized Amazon Cognito and SQL to implement secure, efficient backend infrastructure.
- Designed and implemented intuitive UI to display certifications, commission card details, and employee ID.

Web Scraper | JavaScript, HTML, CSS (February 2024-March 2024)

- Constructed JavaScript-based scraper to parse Ohio State class schedule website.
- Parsed for specific credit hours and weekday meeting schedules.
- Implemented output function for console display, text file export, and CSV formatting.
- Designed HTML webpage with CSS styling to display collected data in user-friendly format.

Professional Experience

CONSTRUCTCONNECT – Cincinnati, OH

June 2025 to Present Day

Construction software company providing AI-powered project takeoff tools, connected project data, and bid management solutions to help subcontractors find the right projects, bid faster, and win more work through integrated workflows.

Automation Developer

Develop and maintain automated workflows and microservices within Content division to streamline construction project data processing and publishing. Design Python-based automation solutions that integrate with APIs, databases, and Microsoft products to optimize business processes and enhance efficiency for content teams. Collaborate with stakeholders to identify automation opportunities and implement AI-driven solutions for data transformation and system integration.

- **Engineered project upload microservice in TypeScript** to automatically post construction project data (titles, locations, event dates, structures, contacts) to UCMS API, reducing manual data entry time for content team members.
- Developed contact matching microservice to **automatically match construction project contacts to existing company and contact IDs in UCMS database**, enabling accurate contact attribution for owners, architects, and contract administrators.
- Conducted **extensive testing and debugging** of microservices to improve test coverage and optimize performance.

FEDERAL HOME LOAN BANK OF CINCINNATI (FHLB) – Cincinnati, OH

May 2024 to August 2024

Regional wholesale bank providing financial services for residential housing and economic development to 603 member-stockholders in FHLB System's Fifth District of Kentucky, Ohio and Tennessee.

Software Engineering Intern

Collaborated across multiple scrum teams to support internal software development initiatives in housing and finance departments. Assisted debugging team in identifying and resolving defects within mission-critical applications used across internal systems. Rotated through Agile sprints to expand understanding of various department functions and workflows. Gained hands-on experience with Azure Cloud services and Agile software development processes.

- Contributed to full development lifecycle by **building features, debugging legacy code, and writing test cases in C# and Angular environments with NgRx integration**.
- Improved performance of data-driven systems by **updating and deploying optimized SQL queries**.
- **Supported internal development efforts by contributing to automation of UAT (user acceptance testing)**.

Technical Skills

Programming Languages & Scripting:	Python, Java, C, C++, C#, JavaScript, TypeScript, HTML, CSS, SQL, Ruby, Bash
Frameworks & Development Tools:	Angular, Rails, NgRx, Expo, React Native, Visual Studio, Visual Studio Code, IntelliJ, Docker
Cloud & Database Technologies:	Azure, AWS, GCP, Amazon Cognito, SQL, SSMS
Version Control & Collaboration:	Git, GitHub, GitLab, Azure DevOps
Game & Interactive Media Development:	Unity, C#, Prim's Algorithm
Certifications & Technical Knowledge:	Microsoft Office Specialist – Word, Excel, PowerPoint, Access, Outlook, Hardware assembly, technical troubleshooting, internet proficiency, email proficiency, Data Structures & Algorithms, Cybersecurity, Software Development, Computer Networking, Software Engineering Techniques

Volunteering/Community Service

Eagle Scout, Boy Scouts of America – Troop 500 | 2012–2018

Patrol Leader, Troop Quartermaster, Troop Guide, and Troop Instructor

- Completed 21 merit badges to achieve Eagle Scout rank and demonstrate long-term commitment. Received Bronze Palm Award by completing 5 additional merit badges beyond rank requirements.
- Elected as member of The Order of the Arrow National Honor Society for embodying Scout Oath and Law.
- Planned and led service project constructing 10 wooden picnic tables for Fort Ancient Earthworks and Nature Preserve.